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Formal versus Informal Finance: Evidence from China

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1 Big picture

- **Question.** In a fast-growing but financially underdeveloped economy like China, is firm growth mainly supported by **informal finance** and relationship-based mechanisms, or does **formal bank finance** still matter more for firm performance?
- **Setting.** The paper studies about 2,400 Chinese firms from the 2003 World Bank Investment Climate Survey, with quantitative firm information for 1999–2002 and firms drawn from 18 cities across five regions.
- **Core challenge.** Financing choice is endogenous. Fast-growing firms may be more likely to seek or obtain bank loans, and banks may lend selectively to firms with better collateral or private information about prospects. So simple correlations between financing and growth are not enough.
- **Main empirical result.** Firms with bank financing grow faster and reinvest more profits than otherwise similar firms. External informal financing is **not** associated with faster sales growth, and it is often associated with **lower** reinvestment.
- **Main nuance.** If “informal” financing is defined broadly to include **retained earnings**, then self-financing can look beneficial. But this positive association is driven mainly by **internal funds**, not by moneylenders or other external informal channels.
- **Additional result.** The positive association between bank finance and firm performance survives a treatment-effects model, IV estimation, and matching exercises. Firms that receive government help in obtaining bank loans do not perform better than other bank-financed firms.
- **Why this paper matters.** It pushes back against the view that China’s private-sector growth is mainly explained by informal finance as a substitute for formal finance. The paper argues instead that, even in an imperfect banking system, bank finance is the channel most strongly associated with better firm outcomes.

2 Data and measurement (key objects)

2.1 Observed firm-level panel

- **Source.** The World Bank Investment Climate Survey for China, implemented in early 2003 with the Chinese National Bureau of Statistics.
- **Coverage.** 2,400 firms from 18 cities in five regions: Northeast, Coastal, Central, Southwest, and Northwest.
- **Industries.** Both manufacturing and services are included. The manufacturing industries include apparel and leather goods, electronics, consumer products, vehicles, and vehicle parts; the services industries include accounting, advertising, logistics, communications, and IT services.
- **Size coverage.** The survey is unusually rich in small and medium enterprises. Firms with fewer than 300 employees are 72% of the sample, medium firms (300–2,000 employees) are 21.2%, and large firms are 6.7%.

- **Time dimension.** Most qualitative questions refer to 2002, but a short quantitative panel exists from 1999 to 2002.
- **Ownership and organization.** The survey records whether firms are corporations, state-owned enterprises, cooperatives, or other legal forms, and it provides ownership breakdowns into domestic/private, state, and foreign ownership.

2.2 Financing measures

Firms report the shares of **new investment financing** and **working-capital financing** coming from:

- local commercial banks,
- foreign commercial banks,
- trade credit,
- investment funds / special development finance / state services,
- retained earnings,
- loans from family and friends,
- informal sources such as moneylenders or informal banks,
- equity,
- other sources.

The paper groups these sources in two ways.

- **Six-way categorization.** Bank financing, informal financing, operations financing, equity financing, investment funds, and internal financing.
- **Allen–Qian–Qian categorization.** Bank financing versus self-fund raising, where self-fund raising is everything except bank finance.

The main bank-access variables are:

- **Bank Dummy:** equals 1 if the firm reports a current bank loan and an overdraft / line of credit, and 0 if it reports neither.
- **Access Dummy:** equals 1 if the firm had access to a bank loan at any point from 1990 to 2001.
- **Bank Financing Dummy:** equals 1 if the firm both has bank credit and reports that bank financing covers at least 50% of either new investment or working capital; equals 0 if it has no bank credit and bank financing of both is zero.

The two self-financing variables are:

- **Self-Financing1:** equals 1 if more than 50% of new investment or working capital is financed by **Informal + Other Sources**; equals 0 if both are zero.
- **Self-Financing2:** equals 1 if more than 50% is financed by **Retained Earnings + Family and Friends + Informal + Other**; equals 0 if these are all zero.

2.3 Firm performance and controls

The main dependent variables are:

- **Sales Growth** from 2001 to 2002:

$$\text{Sales Growth}_{i,2001-2002} = \log(\text{Sales}_{i,2002}) - \log(\text{Sales}_{i,2001}).$$

- **Sales Growth** from 1999 to 2002, used as a longer-horizon robustness outcome.
- **Labor Productivity Growth**, where labor productivity is

$$LP_{it} = \frac{\text{Sales}_{it} - \text{MaterialCosts}_{it}}{\text{Workers}_{it}},$$

and growth is the log change in this measure.

- **Reinvestment Rate**, defined as the share of net profits reinvested in the firm rather than distributed.
- **Growth in TFP**, computed as the residual-based log change in total factor productivity between 2001 and 2002 using value added, labor, and fixed assets interacted with industry dummies.

Average outcomes in the sample are economically meaningful: one-year sales growth is 5.6%, three-year sales growth is 13.5%, one-year labor-productivity growth is 0.45%, three-year labor-productivity growth is 8.1%, and the reinvestment rate averages 17.6%.

The controls include:

- size quintile dummies based on 1999 sales,
- age dummies (less than 5 years, 5–20 years, more than 20 years),
- corporation and cooperative / collective legal-form dummies,
- a state-ownership dummy,
- competition dummies based on the number of domestic competitors,
- city fixed effects, and in robustness checks industry dummies.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline reduced-form specification

The starting point is the baseline regression

$$\begin{aligned} \text{Performance}_i = & \alpha + \beta_1 \text{Bank Dummy}_i + \beta_2 \text{Size Dummies}_i + \beta_3 \text{Age Dummies}_i \\ & + \beta_4 \text{Corporation Dummy}_i + \beta_5 \text{Collectives Dummy}_i + \beta_6 \text{State Ownership Dummy}_i \\ & + \beta_7 \text{Competition Dummies}_i + \beta_8 \text{City Dummies}_i + \varepsilon_i. \end{aligned} \tag{1}$$

- **Interpretation.** This is a reduced-form association between formal bank finance and firm performance.
- **Why it is not enough.** The coefficient on bank finance may be biased by self-selection into loans or by bank screening.
- **Why it still matters.** Even absent a causal interpretation, a positive coefficient says that fast-growing firms reveal a preference for bank finance over informal finance at the margin.

3.2 Treatment-effects logic and the ATT

The paper frames the problem using the average treatment effect on the treated. Let Y_i^1 be the firm's performance if bank financed and Y_i^0 the performance if not bank financed. Then the object of interest is

$$\tau \mid \text{Bank} = 1 = E(Y_i^1 \mid \text{Bank}_i = 1) - E(Y_i^0 \mid \text{Bank}_i = 1). \quad (2)$$

Because $E(Y_i^0 \mid \text{Bank}_i = 1)$ is unobserved, the naive comparison is

$$\tau = E(Y_i^1 \mid \text{Bank}_i = 1) - E(Y_i^0 \mid \text{Bank}_i = 0), \quad (3)$$

which is biased if firms that get loans are systematically different from those that do not.

- **Interpretation.** The econometric problem is classic sample selection: treated firms are not randomly assigned.
- **Economic meaning.** Banks may screen on information the econometrician does not observe, such as borrower quality or collateral quality.

3.3 Selection equation and exclusion restriction

The paper uses a Heckman-style two-step selection model. Bank access is modeled as

$$\begin{aligned} \text{Bank Dummy}_i = 1 \quad \text{if} \quad & \alpha_0 + \beta_1 \text{Collateral}_i + \beta_2 \text{Size Dummies}_i + \beta_3 \text{Age Dummies}_i \\ & + \beta_4 \text{Corporations}_i + \beta_5 \text{Collectives}_i + \beta_6 \text{State Ownership}_i \\ & + \beta_7 \text{Competition Dummies}_i + \beta_8 \text{City Dummies}_i + \zeta_i > 0. \end{aligned} \quad (4)$$

The second-stage performance equation is

$$\begin{aligned} \text{Performance}_i = & \alpha_1 + \gamma_1 \text{Bank Dummy}_i + \gamma_2 \text{Size Dummies}_i + \gamma_3 \text{Age Dummies}_i \\ & + \gamma_4 \text{Corporations}_i + \gamma_5 \text{Collectives}_i + \gamma_6 \text{State Ownership}_i \\ & + \gamma_7 \text{Competition Dummies}_i + \gamma_8 \text{City Dummies}_i + \lambda_i + \varepsilon_i, \end{aligned} \quad (5)$$

where λ_i is the inverse Mills ratio from the first-stage selection equation.

- **Key instrument.** **Collateral**, a dummy that proxies the firm's ability to pledge collateral acceptable to banks.
- **Economic rationale.** Chinese banks rely heavily on land and buildings as collateral, and lack of collateral is the main reason that loan applications are rejected.
- **Identification logic.** Collateral affects access to bank loans but does not directly affect short-run growth once other firm characteristics are controlled for.

3.4 Alternative estimators and robustness designs

Beyond the selection model, the paper also uses:

- **LIML-IV regressions**, where Bank Dummy is instrumented with Collateral, Government Help, or both;
- **Propensity-score matching**, to compare bank-financed firms with similar non-bank-financed firms;
- **Alternative finance-share regressions**, using the financing composition of new investment and working capital;
- **Subsample checks**, including private firms only, dropping firms with government help, and examining firms that did not apply for loans.
- **Interpretation**. The paper does not rest on one single estimator. It layers several approaches to show that the bank-finance result is robust.
- **Important nuance**. Because financing shares are only observed for 2002, those regressions are treated more as consistency checks than as clean causal estimates.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the simple correlation approach

- **Reverse causality**. Fast-growing firms may be more likely to seek bank finance, so a positive bank-growth correlation need not imply causality.
- **Selection into application**. Many firms never apply for a loan, so comparing firms with and without loans may confound loan access with self-selection.
- **Bank private information**. Banks may observe borrower quality not seen by the econometrician.
- **Directed credit concern**. If government pressure directs banks toward particular firms, estimated bank-finance effects could partly reflect politically selected borrowers rather than bank finance itself.

4.2 What identifies the paper's preferred strategy

- **Collateral as exclusion restriction**. Since Chinese banks require collateral—mainly land and buildings—collateral availability shifts loan access in a way that is plausibly orthogonal to short-run growth shocks after controls.
- **Government Help as an additional test**. The paper uses a government-help dummy both as a direct control and as an instrument in robustness exercises to ask whether political allocation is driving the results.
- **Nonapplicant and rejected-applicant evidence**. Firms that do not apply for loans or are rejected do not outperform bank-financed firms, which weakens the argument that informal-finance users are simply better firms that do not need banks.

- **Matching evidence.** Propensity-score matching yields the same qualitative result: bank-financed firms grow faster and reinvest more than comparable firms.

4.3 Relation to the literature

- **Against the “China exception” view.** The paper directly challenges Allen, Qian, and Qian (2005), who emphasize self-fund raising and relationship-based mechanisms as explanations for Chinese private-sector growth.
- **Link to finance and growth.** The findings are much closer to the mainstream finance-growth literature: formal finance remains important even in an economy with weak legal and financial institutions.
- **Nuanced view of informal finance.** Informal finance may still matter for tiny firms with limited bank access, much like angel finance for young firms in developed economies, but it does not scale as a broad substitute for banks.

5 Tables and figures

5.1 Table 1: Financing patterns in China compared to other countries

Table 1 continued Panel B: Aggregate financing patterns												Table 1 continued Panel B: Aggregate financing patterns												Allen et al. (2005) categorization							
												Aggregate financing patterns																			
												Number of countries		Number of firms		Internal Financing		Bank Financing		Informal Financing		Operations Financing		Equity Financing		Investment Funds		Bank Financing		Self-Fund Raising	

Read:

- China is *not* an anomaly in its aggregate reliance on nonbank finance once financing sources are grouped broadly. In the Allen–Qian–Qian classification, Chinese firms finance about **20.37%** of new investment through banks and **79.63%** through self-fund raising.
- That bank-financing share is similar to Africa (about 19%), South Asia (about 23%), and Latin America and the Caribbean (about 21%). So the paper’s first message is: China’s firms do not look extraordinary in aggregate bank usage.
- What *is* unusual is the composition of nonbank finance. China has comparatively low retained earnings financing (**15.24%**) and a very large “Other Sources” category (**42.70%**), plus relatively high equity financing (**12.39%**).
- The paper therefore accepts that China uses many alternative channels, but it rejects the stronger claim that this makes China fundamentally different from other developing economies in overall reliance on formal finance.

5.2 Table 2: Financing patterns across Chinese regions and firm types

Table 2
Financing patterns in China

Panel A: Individual financing patterns

	Number of firms	Retained Earnings	Local Commercial Banks	Foreign-Owned Commercial Banks	Operations Finance	Family and Friends	Informal Sources	Equity	Investment Funds	Other
Across regions in China										
Central	379	16.61	18.18	0.05	0.34	7.62	2.57	8.26	0.17	46.18
Coastal	265	15.27	23.19	0.47	1.46	3.39	0.34	16.49	0.49	38.89
Northwest	296	13.38	18.46	0.00	0.84	4.34	1.97	12.62	1.05	47.33
Northwest	89	13.07	14.15	0.00	1.12	6.29	1.78	11.42	0.80	51.38
Southwest	258	15.78	25.78	0.00	1.78	5.13	2.20	14.33	0.07	50.02
Across all firms	1,287	15.16	20.52	0.11	1.03	5.41	1.85	12.39	0.46	43.07
Across firm sizes										
Micro	250	11.41	15.06	0.00	0.48	13.54	2.58	12.78	0.84	43.31
Small	249	14.20	14.82	0.04	1.14	6.70	1.59	11.97	0.10	47.45
Medium	249	16.50	21.89	0.46	0.60	4.62	1.49	11.14	0.94	42.56
Large	249	17.08	21.35	0.00	1.10	2.79	1.99	12.14	0.14	45.45
Very Large	249	17.02	30.13	0.08	1.61	0.08	1.81	11.25	0.29	50.21
Legal status										
Publicly listed company	37	9.00	33.32	0.00	0.00	0.00	0.00	36.11	0.00	21.57
Privately held company	490	18.11	20.54	0.03	0.98	10.00	2.21	14.14	0.26	35.75
Cooperatives/Collections	165	15.92	17.59	0.00	0.55	3.33	0.99	10.65	0.06	48.92
Across firm ownership (1-50%)										
State	288	11.69	25.90	0.00	0.02	0.79	1.47	6.36	0.59	53.18
Domestic private	901	16.23	19.60	0.14	1.10	7.30	2.00	14.16	0.36	39.12
Foreign private	108	14.56	14.56	0.19	3.36	1.07	1.39	13.64	0.93	48.08
Age										
Less than 3 years	362	14.83	20.53	0.00	0.73	10.30	2.61	16.33	0.53	34.18
Between 3 and 20 years	735	16.67	18.16	0.20	1.45	4.85	1.52	13.53	0.62	43.01
Greater than 20 years	290	11.67	26.51	0.00	0.24	2.40	1.99	5.96	0.00	51.23

Table 2
continued
Panel B: Aggregate financing patterns

	Number of firms	Internal Financing	Bank Financing	Aggregate financing patterns Informal Financing	Operations Financing	Equity Financing	Investment Funds	Bank Financing	Self-Fund Raising
Across regions in China									
Central	379	70.41	18.23	0.34	8.26	0.17	18.23	81.77	
Coastal	265	57.55	23.66	0.34	1.46	16.49	0.49	23.66	76.34
Northwest	296	65.05	18.46	1.97	0.84	12.62	1.05	18.46	81.54
Northwest	89	70.74	14.15	1.78	1.12	11.42	0.80	14.15	85.85
Southwest	258	55.85	25.78	2.20	1.78	14.33	0.07	25.78	74.22
Across all firms	1287	63.64	20.63	1.85	1.03	12.39	0.46	20.63	79.37
Across firm sizes									
Micro	250	68.27	15.06	2.58	0.48	12.78	0.84	15.06	84.94
Small	249	68.35	14.86	1.59	1.14	13.97	0.10	14.86	85.14
Medium	249	63.67	22.25	1.49	0.60	11.14	0.94	22.25	77.65
Large	249	63.27	21.35	1.99	1.10	12.14	0.14	21.35	78.65
Very Large	249	54.84	30.21	1.81	1.61	11.25	0.29	30.21	69.79
Legal status									
Publicly listed company	37	30.57	33.32	0.00	0.00	36.11	0.00	33.32	66.68
Privately held company	490	61.86	20.57	2.21	0.98	14.14	0.26	20.57	79.43
Cooperatives	165	70.16	17.59	0.99	0.55	10.65	0.06	17.59	82.41
Across firm ownership									
Domestic private	288	65.66	25.90	1.47	0.02	6.36	0.59	25.90	74.10
Foreign private	901	62.65	19.74	2.00	1.10	14.16	0.36	19.74	80.26
Foreign private	108	65.94	14.75	1.39	3.36	13.64	0.93	14.75	85.23
Age									
Less than 5 years	262	59.29	20.53	2.61	0.73	6.36	0.59	20.53	79.47
Between 5 and 20 years	735	64.53	18.35	1.52	1.45	14.16	0.36	18.35	81.65
Greater than 20 years	290	65.30	26.51	1.99	0.24	3.64	0.93	26.51	73.49

This table presents financing patterns across different regions in China and across different types of firms. In Panel A, Retained Earnings, Local Commercial Banks, Foreign-Owned Commercial Banks, Equity, Operations Finance, Investment Funds, Loans from Family and Friends, Informal Sources (e.g., moneylenders), and Other Sources are financing proportions stand for the proportion of new investments financed by each of these sources. Operations Finance consists of financing from leasing, trade credit, and credit cards. Investment Funds includes state fund investment funds, development bank, and other state-owned financial institutions. In Panel B, we present two types of aggregate financing patterns: Bank Financing (Bank Loans and Earnings, Loans from Family and Friends, and Other Sources), Bank Financing (Local Commercial Banks and Foreign-Owned Commercial Banks), Informal Financing (Informal sources as moneylender or informal bank), Equity Financing, Operations Financing, and Investment Funds. The second set consists of Bank Financing and Self-Fund Raising (100-Bank Financing) as defined in Allen, Qian, and Qian (2005) to represent the proportion of new investments financed by all other sources except Bank Financing. The financing proportions are in percentages.

Read:

- Access to bank financing varies systematically by region. Aggregate bank financing is highest in the **Southwest (25.78%)** and **Coastal region (23.66%)**, and lowest in the **Northwest (14.15%)**.
- Bank finance rises strongly with firm size. Micro and small firms receive only about **15%** bank financing, whereas very large firms receive about **30.21%**.
- Legal form and ownership matter too. Publicly listed firms finance about **33.32%** of new investment through banks, compared with **17.59%** for cooperatives. State-owned firms use about **25.90%** bank financing, while domestic private firms use about **19.74%**.
- Older firms rely more on banks than younger firms. Firms older than 20 years use about **26.51%** bank financing, while the youngest firms rely much more heavily on self-fund raising.

5.3 Table 3: Summary statistics and simple correlations

Table 3
Summary statistics and correlations

Panel A: Summary statistics					
Variable	N	Mean	Standard deviation	Minimum	Maximum
Financing					
Bank Dummy	2,326	0.2309	0.4215	0	1
Access Dummy	2,400	0.2858	0.4518	0	1
Bank Financing	941	0.2721	0.4452	0	1
Self-Financing1	1,446	0.6023	0.4896	0	1
Self-Financing2	1,502	0.8302	0.3756	0	1
Firm Performance					
Sales Growth [2001-2002]	2,370	0.0563	0.3072	-7.44	7.13
Reinvestment Rate [2001-2002]	2,115	0.1761	0.3230	0	1
Sales Growth [1999-2002]	2,259	0.1450	0.3060	-2.36	2.71
Labor Productivity Growth [2001-2002]	1,358	0.0045	0.8132	-6.68	8.62
Labor Productivity Growth [1999-2002]	1,486	0.0010	0.3397	-1.97	3.28
Growth in TFP [2001-2002]	693	-0.0192	0.9521	-4.84	6.31
Firm Characteristics					
Size Dummies	2,283	2.9991	1.1445	1	5
Age	2,400	15.9862	14.2962	3	55
Corporation Dummy	2,400	0.4046	0.4909	0	1
Cooperatives/Collections Dummy	2,400	0.0162	0.3678	0	1
State Ownership Dummy	2,399	0.2213	0.4152	0	1
Competition Dummies	2,326	3.8224	1.3535	1	5

Table 3
Continued

Panel B: Correlations between financing and firm performance

	Sales Growth [2001-2002]	Reinvestment Rate	Labor Productivity Growth [2001-2002]	Sales Growth [1999-2002]	Labor Productivity Growth [1999-2002]	Bank Dummy	Access Dummy	Bank Financing	Self-Financing1
Reinvestment Rate	0.016***	-0.0111	0.630***	0.1171***	0.2982***				
Sales Growth [1999-2002]	0.5072***	0.4218***	0.0484**	0.4602***	0.7298***				
Labor Productivity Growth [1999-2002]	0.0337*	0.132***	0.0058	0.0671***	0.0573**				
Access Dummy	0.0003	-0.0076	-0.0332	-0.061***	-0.0601**	0.0014***			
Bank Financing	0.0366	0.1664***	0.0279	0.024***	0.0878***	1***	0.0762***		
Self-Financing1	-0.0077	-0.0023***	0.0639***	-0.0401**	0.0016	-0.1174***	-0.0844***	-0.1362***	
Self-Financing2	0.0009	-0.0672***	0.0746***	-0.0229	0.0111	-0.1281***	-0.0801***	-0.1061***	0.9301***

*Significant at 10%; **significant at 5%; ***significant at 1%.

Panel A presents summary statistics. Panel B presents the correlations between the financing variables and firm performance. Bank Dummy takes the value of 1 if the firm states that it has a current loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said it had no bank loan and had no overdraft facility or line of credit. Access Dummy is a dummy variable that takes the value of 1 if the firm had access to a bank loan in any year prior from 1990-2001 and 0 otherwise. Bank Financing Dummy takes the value of 1 if the firm states that it has a loan from a bank or financial institution and an overdraft facility or line of credit and reports that the bank finances at least 50% of new investments or working capital. Bank Financing takes the value of 0 if the firm states that it has no loan from a bank or financial institution and said it had no overdraft facility or line of credit and bank financing of both new investments and working capital was equal to 0%. Self-Financing1 takes the value of 1 if the sum of internal financing and Other Financing of other new investments or working capital is greater than 50%. Self-Financing2 takes the value of 0 if the sum of informal and other financing of new investments and working capital is equal to 0%. Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002 and from 1999 to 2002. Labor Productivity Growth is defined as the log change in labor productivity where labor productivity is Sales / Total Material Costs/Total Number of Workers. Productivity Growth is computed from 2001 to 2002 and from 1999 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Firm Size Dummies are quintiles of total firm sales in 1999. Age is the age of the company in 2003. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperatives/Collections Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. Competition takes values 1 to 5 for 1-3 competitors, 4-6 competitors, 7-10 competitors, and over 100 competitors, respectively, for the number of competitors in its main business line in the domestic market.

Read:

- Only a minority of firms have meaningful access to the formal banking system. The mean of **Bank Dummy** is **0.2309**, and the mean of **Access Dummy** is **0.2858**.
- Out of 2,326 firms answering the current-loan question, only **537** report having a bank loan. Among firms without a current loan, **1,237** did not apply and **229** were rejected.

- Collateral is the central barrier to formal finance. Of the 229 rejected applicants, about **66%** cite lack of collateral as the main reason their application was rejected.
- The raw correlations already point in the paper's direction: bank-finance measures are positively associated with growth outcomes, while self-financing measures are negatively associated with bank-finance access and do not show a robust positive relation with sales growth.

5.4 Table 4: Baseline regressions of bank finance on firm performance

Table 4
Bank financing and firm performance—partial correlations

Panel A: Bank Dummy	Full sample						Only public corporations and state-owned companies					
	1	2	3	4	5	6	7	8	9	10	11	12
	Bank Dummy	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned	Bank Dummy	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned
Bank Dummy	1.000	0.078***	0.078***	0.078***	0.078***	0.078***	1.000	0.078***	0.078***	0.078***	0.078***	0.078***
Sales		1.000	0.078***	0.078***	0.078***	0.078***		1.000	0.078***	0.078***	0.078***	0.078***
Productivity			1.000	0.078***	0.078***	0.078***			1.000	0.078***	0.078***	0.078***
Reinvestment				1.000	0.078***	0.078***				1.000	0.078***	0.078***
Bank Financing					1.000	0.078***					1.000	0.078***
State-Owned						1.000						1.000

Table 4
Continued

Panel B: Alternative definitions of bank finance	1	2	3	4	5	6	7	8	9	10	11	12
	Bank Dummy	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned	Bank Dummy	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned
	1.000	0.078***	0.078***	0.078***	0.078***	0.078***	1.000	0.078***	0.078***	0.078***	0.078***	0.078***
Observation	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222
R-squared	0.006	0.072	0.072	0.072	0.072	0.072	0.006	0.072	0.072	0.072	0.072	0.072

Read:

- In the full sample, **Bank Dummy** is associated with about **7.5 log points** higher sales growth over 2001–2002 and about **11.5 log points** higher sales growth over 1999–2002.
- Bank finance is also associated with materially higher reinvestment: the coefficient on the reinvestment rate is about **0.078** in the full sample and **0.086** in the private-firm subsample.
- The key discipline is that bank-financed growth does *not* come with worse efficiency. The coefficients on one-year labor-productivity growth and TFP growth are positive or near zero and statistically insignificant, not negative.
- The private-firm sample tells the same story, and that is important: even among the firms often argued to rely on informal channels, bank finance is the variable associated with stronger growth.

5.5 Table 5: Treatment-effects / selection model

Read:

- Once the paper controls for selection into bank finance, the estimated bank-finance effects become **larger**, not smaller. In the full sample, the bank-finance coefficient rises to about **0.310** for sales

Table 4
Continued

Bank Financing of Working Capital (Continuous)	1	2	3	4	5	6	7	8	9	10	11	12
	Bank Financing of Working Capital	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned	Bank Financing of Working Capital	Sales Growth	Productivity Growth	Reinvestment Rate	Bank Financing	State-Owned
	1.000	0.078***	0.078***	0.078***	0.078***	0.078***	1.000	0.078***	0.078***	0.078***	0.078***	0.078***
Observation	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222	2,222
R-squared	0.006	0.072	0.072	0.072	0.072	0.072	0.006	0.072	0.072	0.072	0.072	0.072

*Significant at 10%. **Significant at 5%. ***Significant at 1%.
The estimated model in Panel A is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$ Bank Dummy + β_2 Small + β_3 Medium + β_4 Large + β_5 Very Large + β_6 Mid-Age + β_7 Old + β_8 Cooperatives + β_9 Collectives + β_{10} State Ownership + β_{11} Competition Dummies + β_{12} City Dummies. The estimated model in Panel B is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$ Bank Financing of New Investments (or Working Capital) + β_2 Dummy for 100% Bank Financing of New Investments (or Working Capital) + β_3 Small + β_4 Medium + β_5 Large + β_6 Very Large + β_7 Mid-Age + β_8 Old + β_9 Cooperatives + β_{10} Collectives + β_{11} State Ownership + β_{12} Competition Dummies + β_{13} City Dummies. Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002 and from 1999 to 2002. Labor Productivity Growth is defined as the log change in productivity where productivity is (Sales - Total Material Costs)/Total Number of Workers. Labor Productivity Growth is computed from 2001 to 2002 and from 1999 to 2002. Growth in Total Factor Productivity (TFP) is computed from 2001–2002 where TFP is calculated as the residual from a regression of value added on labor and fixed assets, allowing for industry-specific coefficients for each variable. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Bank Dummy takes the value of 1 if the firm states that it has a current loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm states that it had no bank loan and had no overdraft facility or line of credit. Bank Financing of New Investments (Working Capital) is a continuous variable that measures the proportion of total financing of new investments (working capital) that is bank financed. The Dummy for 100% Bank Financing of New Investments (Working Capital) takes the value of 1 if the proportion of bank financing in total financing of new investments (working capital) is 100% and 0 otherwise. Firm Size Dummies are quintiles of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. The mid-age dummy is less than five years. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperative/Collective Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 7–15 Competitors, 16–100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. Columns (1)–(6) in Panel A and columns (1)–(12) in Panel B present results for the full sample. Columns (7)–(12) in Panel A present results for a sample of firms that do not include firms registered as publicly traded firms and state-owned enterprises. We use OLS regressions with robust standard errors.

Table 5
Bank financing and firm performance—selection model

	Full sample				Drop public corporations and state-owned companies			
	1	2	3	4	5	6	7	8
Bank Dummy	Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002		Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002	
	0.110*** [0.036]	0.183*** [0.050]	0.159*** [0.045]	0.115*** [0.041]	0.244*** [0.042]	0.159*** [0.041]	0.183*** [0.041]	0.159*** [0.041]
Collateral	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]	0.060*** [0.020]
Small	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Medium	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Large	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Very Large	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Mid-Age	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Old	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
Cooperatives	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
State	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
4-6 Competitors	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]
7-15 Competitors	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]	0.022*** [0.007]

growth and **0.183** for reinvestment.

- In the private-firm subsample, the corresponding treatment-effects estimates remain strong: about **0.244** for sales growth and **0.156** for reinvestment.
- Productivity growth remains nonnegative and statistically weak, which reinforces the paper's claim that bank-financed growth is not visibly inefficient growth.
- The negative hazard lambda suggests that firms selected into bank borrowing would otherwise have performed worse than those that were not, so correcting for selection strengthens the positive role of bank finance.

5.6 Table 6: Robustness—access dummy, government help, and nonapplicants

Table 6
Bank financing and firm performance—robustness

	Full sample				Drop public corporations and state-owned companies			
	1	2	3	4	5	6	7	8
Access Dummy	Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002		Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002	
	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]
Collateral	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]
Hazard Lambda	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]
Observations	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1,566	1,566	1,413	1,413	1,102	1,102	1,102	1,102
Panel B: Government help	OLS				OLS			
	Drop firms that report government help				Drop firms that report government help			
Bank Dummy	Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002		Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002	
	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]	0.100*** [0.038]
Observations	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1,799	1,799	1,549	1,549	1,131	1,131	1,131	1,131
F-squared	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.033	0.033	0.033	0.033	0.033	0.033	0.033	0.033
Durbin-Watson	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1.799	1.799	1.549	1.549	1.131	1.131	1.131	1.131
Chi-Sq Test	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1.799	1.799	1.549	1.549	1.131	1.131	1.131	1.131
First-Stage F-Stat	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1.799	1.799	1.549	1.549	1.131	1.131	1.131	1.131
Overspecification test	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1.799	1.799	1.549	1.549	1.131	1.131	1.131	1.131
Conditional LR (CL p-value)	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1.799	1.799	1.549	1.549	1.131	1.131	1.131	1.131

Read:

- Using the broader **Access Dummy** instead of current bank borrowing yields qualitatively similar treatment-effects results: positive coefficients for sales growth and reinvestment, and a positive but imprecise coefficient for productivity growth.
- Dropping firms that report government help in obtaining loans actually makes the basic OLS bank-finance effect on sales growth **stronger** (about **0.100**).
- In the LIML-IV results, using **Collateral** as the instrument gives a strong positive bank-finance effect on sales growth (about **0.324**); using only **Government Help** does not; using both instruments together again produces a positive and significant estimate (about **0.296**).
- Firms that do *not* apply for bank loans grow **slower** and reinvest **less** than firms that do have bank loans. This directly contradicts the idea that the no-loan firms are systematically the high-growth firms that simply do not need banks.

Table 6
Continued

	Full sample				Drop public corporations and state-owned companies			
	1	2	3	4	5	6	7	8
Access Dummy	Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002		Sales Growth [2001–2002]		Profit Reinvestment Rate in 2002	
	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]	0.144*** [0.040]
Collateral	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]
Hazard Lambda	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]	0.040*** [0.013]
Observations	Selection Equation		Profit Reinvestment Rate in 2002		Selection Equation		Profit Reinvestment Rate in 2002	
	1,566	1,566	1,413	1,413	1,102	1,102	1,102	1,102

*Significant at 10%. **Significant at 5%. ***Significant at 1%.

A two-way fixed-effects model is used to estimate the effect of the endogenous Access Dummy (binary treatment) on Firm Performance. The first step is: Access Dummy = $\alpha_0 + \beta_1 \text{Collateral} + \beta_2 \text{Small} + \beta_3 \text{Medium} + \beta_4 \text{Large} + \beta_5 \text{Very Large} + \beta_6 \text{Mid-Age} + \beta_7 \text{Old} + \beta_8 \text{Cooperatives} + \beta_9 \text{State Ownership} + \beta_{10} \text{City Dummies} + \beta_{11} \text{City Dummies}$. The second step is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha_1 + \beta_{12} \text{Access Dummy} + \beta_{13} \text{Small} + \beta_{14} \text{Medium} + \beta_{15} \text{Large} + \beta_{16} \text{Very Large} + \beta_{17} \text{Mid-Age} + \beta_{18} \text{Old} + \beta_{19} \text{Cooperatives} + \beta_{20} \text{State Ownership} + \beta_{21} \text{City Dummies} + \beta_{22} \text{City Dummies}$.

In Panel A, in column (1), we report OLS regressions similar to column (1) of Table 4 but for a reduced sample of firms, which report no government help in obtaining bank finance. In columns (2)–(4), two-stage Instrumental Variable regressions are used. The first-stage regression is: Access Dummy = $\alpha_0 + \beta_1 \text{Collateral} + \beta_2 \text{Government Help Dummy} + \beta_3 \text{Small} + \beta_4 \text{Medium} + \beta_5 \text{Large} + \beta_6 \text{Very Large} + \beta_7 \text{Mid-Age} + \beta_8 \text{Old} + \beta_9 \text{Cooperatives} + \beta_{10} \text{State Ownership} + \beta_{11} \text{City Dummies} + \beta_{12} \text{City Dummies}$. The second stage is: Sales Growth = $\alpha_1 + \beta_{13} \text{Access Dummy} + \beta_{14} \text{Small} + \beta_{15} \text{Medium} + \beta_{16} \text{Large} + \beta_{17} \text{Very Large} + \beta_{18} \text{Mid-Age} + \beta_{19} \text{Old} + \beta_{20} \text{Cooperatives} + \beta_{21} \text{State Ownership} + \beta_{22} \text{City Dummies} + \beta_{23} \text{City Dummies}$.

In Panel C, we report OLS regressions similar to Table 4 using the following specification: Sales Growth = $\alpha_1 + \beta_2 \text{Did Not Apply for a Bank Loan} + \beta_3 \text{Small} + \beta_4 \text{Medium} + \beta_5 \text{Large} + \beta_6 \text{Very Large} + \beta_7 \text{Mid-Age} + \beta_8 \text{Old} + \beta_9 \text{Cooperatives} + \beta_{10} \text{State Ownership} + \beta_{11} \text{City Dummies} + \beta_{12} \text{City Dummies}$.

Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Access Dummy is a dummy variable that takes the value of 1 if the firm had access to a bank loan in any year prior to 1990 to 2001 and 0 otherwise. Bank Dummy takes the value of 1 if the firm said it had a loan from a bank or financial institution and an overdraft facility or line of credit and 0 if the firm said it had no bank loan and had no overdraft facility or line of credit. The identifying variables in Panel A and columns (2) and (4) of Panel B is Collateral, which is a dummy variable that takes the value of 1 if the financing required a collateral or a deposit and 0 if the financing did not require collateral or if the firm did not apply for a loan because of stringent collateral requirements or if the firm was rejected for a loan because of the lack of collateral. In columns (1) and (4) of Panel B, we also use Government Help dummy as an identifying variable, which takes the value of 1 if a government agency or official assisted the firm in obtaining bank financing. Firm Size Dummies are quintiles of total firm sales in 1999: Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperative/Collective Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4-6 Competitors, 7-15 Competitors, 16-100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1-3 Competitors. Columns (1)–(4) present results for the full sample, and columns (5)–(8) present results for a sample of firms that do not include firms registered as publicly traded firms and state-owned enterprises. Hazard lambda is reported for each of the second stages.

5.7 Table 7: Bank financing shares versus informal/self-financing shares

Table 7
Financing proportions of new investments and working capital—bank vs. informal financing

Panel A: Bank financing						
1	2	3	4	5	6	
Full sample						
Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	
0.062*	0.019**	0.042	0.052	0.074*	0.06	
[0.056]	[0.011]	[0.076]	[0.058]	[0.077]	[0.077]	
895	899	621	659	952	480	
R-squared	0.063	0.099	0.039	0.068	0.103	0.058

Table 7
Continued

Panel B: Self-financing											
1	2	3	4	5	6	7	8	9	10	11	12
Full sample											
Drop public corporations and state-owned companies						Drop public corporations and state-owned companies					
Sales Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Labor Productivity Growth [2001–2002]	Sales Growth [2001–2002]	Sales Growth [2001–2002]	Profit Reinvestment Rate in 2002	Profit Reinvestment Rate in 2002	Labor Productivity Growth [2001–2002]	Labor Productivity Growth [2001–2002]
Self-Financing1 (Informal + Other)	-0.04 [0.039]		-0.126** [0.020]		0.092 [0.039]				-0.127** [0.024]		0.084 [0.064]
Self-Financing2 (Informal + Other + Internal + Loans from Family and Friends)											
		-0.015 [0.048]		0.045* [0.024]		0.154** [0.071]	0.006 [0.059]			0.043 [0.058]	0.151** [0.075]
Observations	1,363	1,372	1,203		952	956	984	992	874	877	719
R-squared	0.056	0.055	0.110	0.103	0.041	0.042	0.06	0.058	0.111	0.048	0.05

*Significant at 10%; **significant at 5%; ***significant at 1%.
The estimated model is: Sales Growth/Reinvestment Rate/Productivity Growth = $\alpha + \beta_1$ Bank Financing or Self-Financing1 or Self-Financing2 + β_2 Small + β_3 Medium + β_4 Large + β_5 Very Large + β_6 Mid-Age + β_7 Old + β_8 Corporations + β_9 Collectives + β_{10} State Ownership + β_{11} City Dummies.
Sales Growth is defined as the log change in total sales and is computed from 2001 to 2002. Reinvestment Rate is the share of net profits reinvested in the establishment in 2002. Bank Financing Dummy takes the value of 1 if the firm states that it has a loan from a bank or financial institution and an overdraft facility or line of credit and reports that bank finances at least 50% of new investments or working capital. Bank Financing takes the value of 0 if the firm states that it has no loan from a bank or financial institution and said it had no overdraft facility or line of credit and bank financing of both new investments and working capital was equal to 0%. Self-Financing1 takes the value of 1 if the sum of financing of either new investments or working capital from Informal and Other Sources is greater than 50%. Self-Financing2 takes the value of 0 if the sum of financing of new investments and working capital from Informal and Other Sources is equal to 0%. Self-Financing2 takes the value of 1 if the sum of financing of new investments or working capital from Retained Earnings, Informal, Loans from Family and Friends, and Other Sources is greater than 50%. Self-Financing2 takes the value of 0 if the sum of financing of new investments or working capital from Retained Earnings, Informal, Loans from Family and Friends, and Other Sources is equal to 0%. Firm Size Dummies are dummies of total firm sales in 1999. Small, Medium, Large, and Very Large dummies take the value of 1 if the firm is in the second, third, fourth, and fifth quintile, respectively, of firm sales. Mid-Age is a dummy variable that takes the value of 1 if the firm is between five and twenty years of age, and Old is a dummy variable that takes the value of 1 if the firm is greater than twenty years old. Corporation Dummy takes the value of 1 if the firm is organized as a corporation (public or private) and 0 otherwise. Cooperative/Collective Dummy takes the value of 1 if the firm is organized as a Cooperative or a Collective. State Ownership Dummy takes the value of 1 if the state owns more than 50% of the company. 4–6 Competitors, 5–15 Competitors, 16–100 Competitors, and Over 100 Competitors are dummy variables that take the value of 1 if the firm has the corresponding number of competitors in its main business line in the domestic market. The omitted category is 1–3 Competitors. We use OLS regressions with robust standard errors.

Read:

- When the paper focuses directly on the financing shares of new investments and working capital, the results still favor formal finance. The **Bank Financing Dummy** is associated with roughly **9.2 log points** higher sales growth and **0.07** higher reinvestment in the full sample.
- By contrast, **Self-Financing1**—which emphasizes *external* informal and other sources—does not predict faster sales growth and is associated with **lower reinvestment** (about **-0.126**).
- Once self-financing is broadened to include **retained earnings** and family funds (**Self-Financing2**), the coefficients on reinvestment and productivity turn positive. This is the paper’s cleanest evidence that the positive story about “informal finance” is really a story about **internal finance**.
- This table is where the paper most directly overturns the Allen–Qian–Qian interpretation: external informal finance is not the high-growth channel; retained earnings are.

6 Short summary

- **Conclusion.** In China, firms with formal bank finance grow faster and reinvest more than similar firms without it. External informal financing is not associated with superior growth.
- **Identification insight.** The paper deals seriously with endogeneity by using collateral-based selection correction, IV, and matching, and these approaches all reinforce the same qualitative result.
- **Main empirical lesson.** The claim that China’s fastest-growing firms are financed mainly by informal channels is too strong. If broad self-financing appears beneficial, it is primarily because **retained earnings** matter, not because moneylenders and other informal financiers outperform banks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that China’s private-sector success is not evidence that formal finance is irrelevant.
- It shows that bank finance is the financing channel most consistently associated with higher sales growth and higher profit reinvestment.
- It also shows that external informal finance does not scale into a general substitute for bank finance. At most, it plays a niche role for the smallest firms.

7.2 Economic intuition

Chinese banks are imperfect, politically influenced, and collateral based. But that does not imply that informal finance works better for most firms. The paper’s evidence suggests a simpler economic story: firms that can access formal bank credit can fund expansion and reinvestment more effectively than firms that rely on informal or miscellaneous outside sources. Informal financiers may be useful for very small or young firms that cannot satisfy collateral requirements, but those arrangements do not seem able to support growth at larger scale. In that sense, informal finance looks more like a lower-end complement than a wholesale substitute for the banking system.

A second intuition is definitional. Once retained earnings are bundled together with informal sources, “informal finance” can appear growth-enhancing. But retained earnings are conceptually different from moneylender credit or underground lending: they are internal funds generated by the firm itself. The paper’s results show that conflating these sources creates the illusion that external informal finance explains Chinese growth.

7.3 Takeaway

This paper is best read as a disciplined empirical rebuttal to the strong “China grows despite banks because of informal finance” claim. Its message is not that China’s banking system is efficient. Rather, it is that even an imperfect formal financial system appears more strongly linked to firm growth and reinvestment than external informal channels. Informal finance may help some micro firms survive, but the scalable, growth-related financing margin in this paper is formal bank finance.